

Intent Based Chatbot

Group Project:-

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Introduction

- What is intent-based chatbot?

A chatbot is a software application that communicates with users through text or voice. An Intent-Based Chatbot identifies the intent (purpose) behind a user's message. It uses Natural Language Processing (NLP) to understand and classify the user's input. Based on the identified intent, the chatbot provides the most appropriate



response. It can be used for customer support, FAQs, information services, and automated assistance.

Problem statement and Proposed solution

- **Problem statement**

Users often have common questions and need quick responses. Providing answers manually can be time-consuming and repetitive. Traditional chatbots may not understand the actual purpose (intent) of a user's message. A system is needed that can understand user queries and provide relevant responses automatically.

- **Proposed solution**

Develop an Intent-Based Chatbot that understands the purpose behind user queries. Use NLP and intent classification to analyze the user's message. Match the detected intent

with the appropriate response. Provide users with fast, accurate, and automated responses.

Objectives

- To develop a chatbot that can understand user queries.
 - To identify the intent behind each user message.
 - To provide relevant and appropriate responses based on the detected intent.
 - To reduce the need for manual assistance for common queries.
 - To create a simple and user-friendly chatbot interface.
- communication.

- To improve the speed and efficiency of

Working & Architecture

- Working Flow

User Input → Text Preprocessing → Intent Classification → Intent Detection → Response Selection → Chatbot Response

- How does it works?
- User Input: The user enters a message or query.
- Text Preprocessing: The input is cleaned and prepared for analysis.
- Intent Classification: The chatbot analyzes the message and predicts its intent.
- Intent Detection: The most suitable intent is identified.
- Response Selection: A response associated with that intent is selected.

- Chatbot Response: The selected response is displayed to the user.

Technologies Used

- **Python**

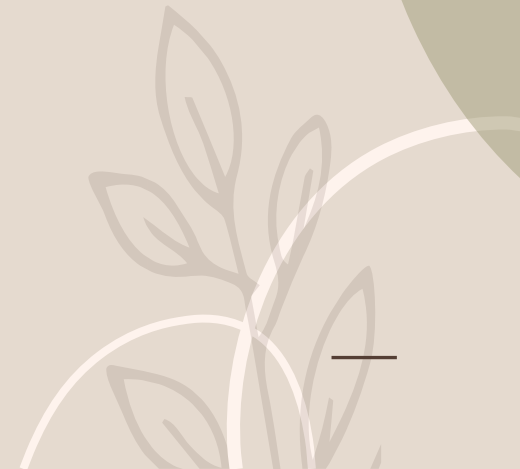
- Main programming language used to develop the chatbot.
- Handles text processing, intent classification, and response generation.

- **NLTK**

- Used for Natural Language Processing (NLP).
- Helps with tasks such as tokenization and text preprocessing.

- **Scikit-learn**

- Used for machine learning and intent classification.
- Helps train the model to identify the intent of user queries.



• Streamlit

- Used to create the interactive chatbot web interface.
- Allows users to communicate with the chatbot through a simple UI

• JSON

- Used to store intents, example user queries, and responses.
- Makes the chatbot's training data easy to manage and modify.

Future Scope and Conclusion

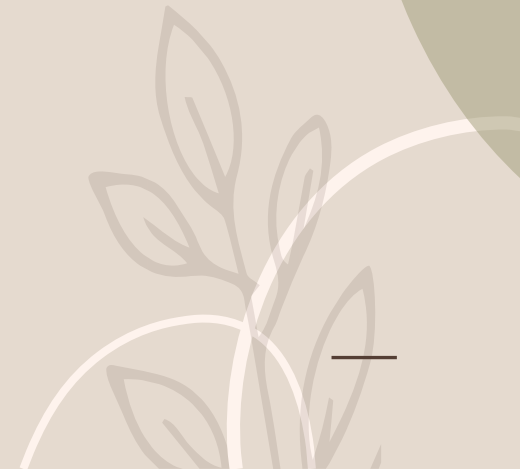
• Future Scope

- Voice-based interaction
- Support for multiple languages
- Better understanding of complex and contextual queries
- Expansion of intents and training data
- Integration with websites and mobile applications
- Use of advanced AI and NLP models

conversational system.

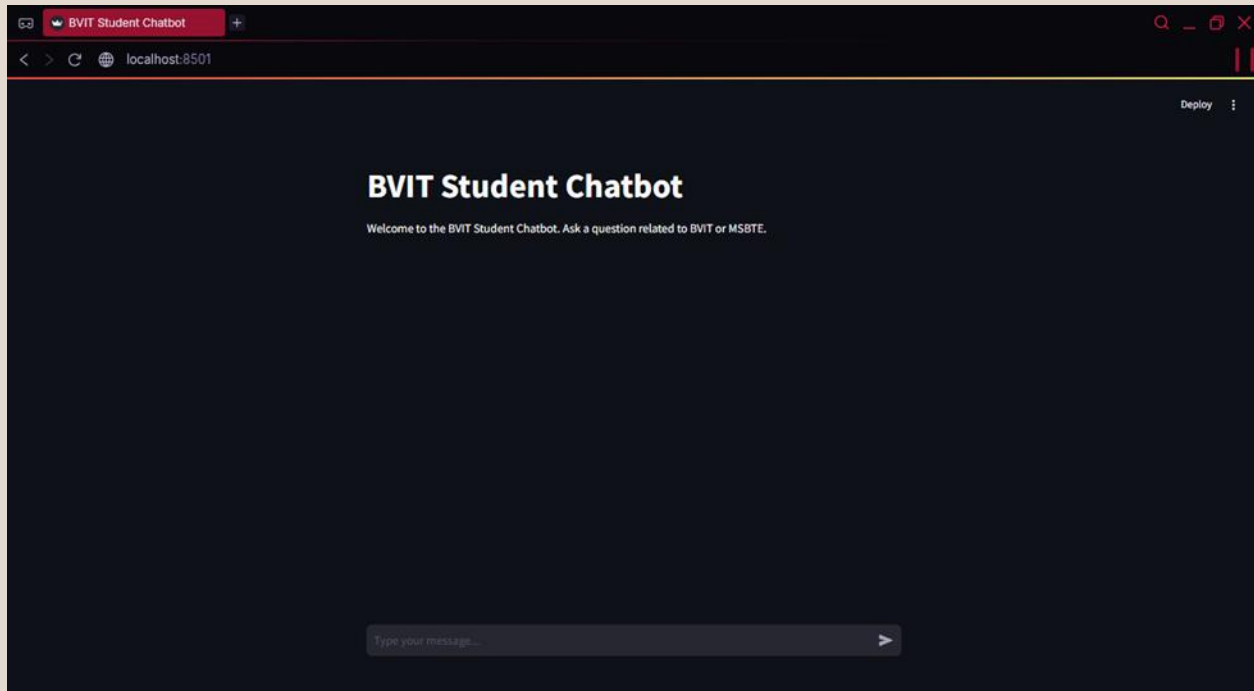
• Conclusion

The Intent-Based Chatbot provides an efficient way to understand user queries and identify their intended purpose. By using Natural Language Processing and

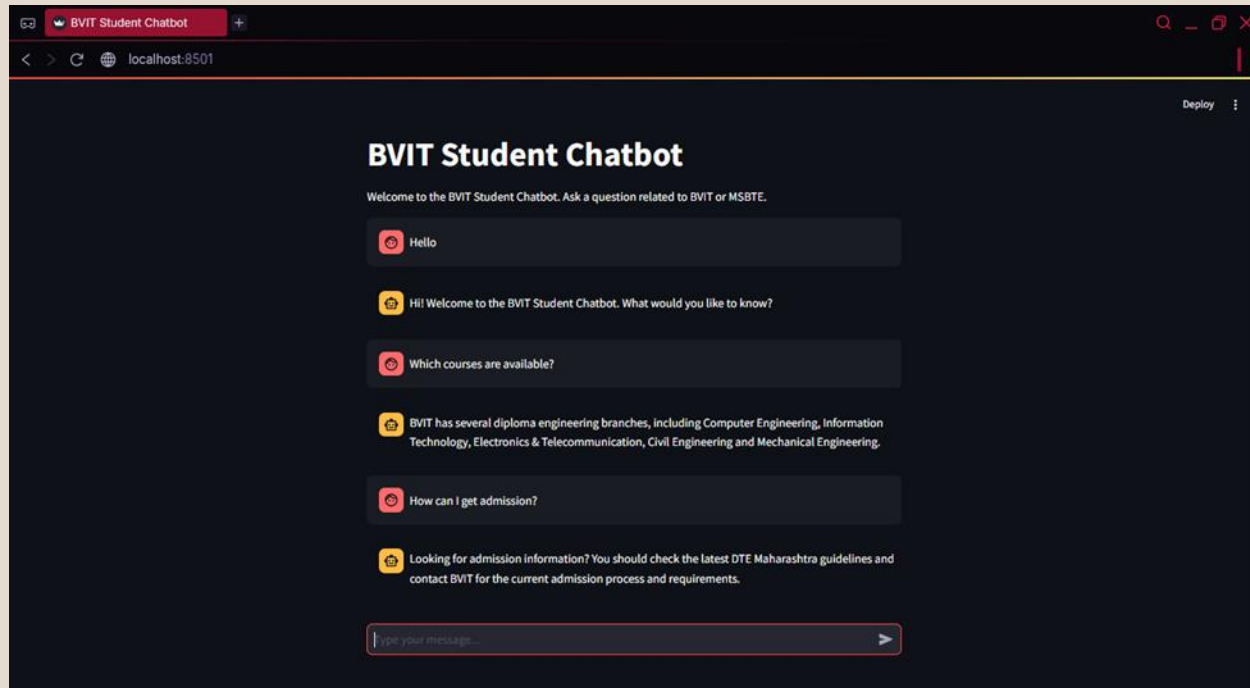


Machine Learning, the chatbot can classify user inputs and provide relevant responses automatically. This project demonstrates how AI and NLP can be used to create a simple, useful, and interactive





Output:





THANK YOU

